

Presentation 1

Links to presentation(s) and code(s) on GitHub

- Link to [Presentation 1](#)
- Link to [Stat 390 Data and Power BI Visuals](#)

What did you do?

Overall Note: While our team collaborated closely, reviewing each other's visuals, sharing feedback, and brainstorming insights together, this documentation focuses on the areas I led and explored most deeply, particularly the development and analysis of redirection-related metrics and visuals.

All Calls Data Cleaning: I processed the All-Calls data into a single CSV file using the import code provided by Prof. Krishna and emailed it to my team. I then loaded this into Power BI to start my analysis. When I opened the file in Power BI, I did the following to clean the data. I focused my data cleaning on the columns I knew I would be using for my visuals.

- I converted 'Called Number' to a text value.
- In the 'Redirect Reason' Column, I Applied Transform → Format → Trim to remove extra spaces and ensure consistent text formatting. I also cleaned up value names and improved readability (ie. changed "NoAnswer" to "No Answer").
- I filtered out all rows where Duration ≤ 0, since calls with zero or negative duration likely represent incomplete or failed call records.

Metric Development: I then started to develop out a metric for redirection efficiency. I came up with looking at **Redirect Success Rate (RSR)** to measure how effectively Legal Aid's phone system redirects incoming calls to the correct destination without unnecessary transfers or failures. I am defining a *redirected call* as referring to any call that, at some point during its lifecycle, was transferred, forwarded, or otherwise routed away from its original destination to another endpoint (such as a queue, another user, or voicemail).

The metric can be calculated as follows:

1. RSR – the percentage of redirected calls that end with a live person answering.
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$$\text{RSR} = \frac{\text{\# of redirected sessions with Success (Human)}}{\text{number of redirected sessions (Correlation IDs)}} \times 100$$

Before defining this metric, I spent time exploring the raw call data and reading through the Webex Calling detailed call-history documentation to understand how different columns interact. I specifically examined the relationships between 'Original reason', 'Redirect reason', 'Related reason', 'Answer indicator', and 'User type' across multiple rows. Through this process, I observed that:

- Calls can contain multiple legs under the same 'Correlation ID', meaning a single call can appear across several rows as it's redirected through the system.
- In some cases, 'Original Reason' and 'Redirect Reason' were blank, yet the Answer indicator showed "Yes-PostRedirection", signaling that the call did experience redirection even if it wasn't recorded in those fields.
- The 'User Type' column helped distinguish between calls answered by a person (e.g., User, CallCenterStandard) and those ending in voicemail or automated systems (e.g., VoiceMailRetrieval, AutomatedAttendantVideo).

I also looked at the [call-number clarification dataset](#) that maps specific phone numbers to their assigned lines or services. From this dataset, it is confirmed that calls ending with 13122296300 represent a human-answered endpoint (front desk). This could also be used in the classification logic for Success (Human) to supplement or replace User type fields with known human-answered numbers where system labeling may be incomplete or ambiguous. However, not all the numbers on this google sheet have descriptions so further work would need to be done on this documentation before using it.

I made the following assumptions based on all of this:

- Blank values for 'Original Reason' mean that no redirection occurred and blank values for 'Redirect Reason' mean no additional redirection occurred
 - This is true unless other evidence (e.g., "Yes-PostRedirection") indicates otherwise
- Each Correlation ID uniquely represents one full call session
- The final answered leg represents the true endpoint of the call
- The User type field reliably distinguishes human endpoints from automated or voicemail systems
- Some of the HuntGroup entries may represent real agents, but this needs confirmation before including them as "human" endpoints

Using these findings, I developed the logic for how to identify when a call qualifies as redirected and how to determine whether it ended successfully. A call session (identified by Correlation ID) is considered redirected if any leg of the call meets one or more of the following conditions:

- Original reason is not blank, or
- Redirect reason is not blank, or
- Related reason indicates redirection activity (e.g., Deflection, CallForward, CallQueue, HuntGroup, SequentialRing, SimultaneousRingPersonal, RoutePoint, Executive), or
- Redirecting number is not blank, or
- Answer indicator = "Yes-PostRedirection"

For success classification:

- Success (Human): The final answered leg (Answered = TRUE) has a User type $\in$ {User, CallCenterStandard} (optionally including HuntGroup if this also represents an agent endpoint).

Once validated, I plan to visualize the RSR metric in Power BI and broken down by Redirect reason (e.g., *NoAnswer*, *CallQueue*, *UserBusy*, *Deflection*, etc.) to highlight where call routing performs well or fails. For example, low RSR for *NoAnswer* may indicate staffing or scheduling gaps or High RSR for *Deflection* suggests that automated attendants or routing rules are working effectively.

Visuals:

After developing the Redirect Success Rate (RSR) metric framework, I created several Power BI visuals to begin exploring the data and validating my assumptions. I also helped write the key insights for all of these listed in the official presentation.

- **Volume of Redirection Events by Redirect Type (Bar Chart)**
 - **X-axis:** Redirect Reason
 - **Y-axis:** Count of Redirect Reason
 - **Filter:** Direction = TERMINATING (includes only inbound, receiving legs)
 - **Purpose:** Displays how frequently each redirection type occurs in the dataset

- **Interpretation Note:** Since each Correlation ID can have multiple call legs, this visual represents the *number of redirection events*, not unique calls
 - Filtering by Direction = TERMINATING ensures the analysis focuses on the *receiving side* of each redirection, where the call actually landed
- **Average Duration per Redirection Type (Bar Chart)**
 - **X-axis:** Redirect Reason
 - **Y-axis:** Average of Duration (seconds)
 - **Filters:** Direction = TERMINATING; Duration > 0
 - **Purpose:** Measures how long calls last once redirected to their final destination.
 - **Interpretation Note:** Duration is measured *per call leg*, not for the entire call session. This helps compare how long interactions last once they reach their redirected endpoints.
- **Recipients of All Calls (Donut Chart)**
 - **Legend:** User Type (e.g., WCCAdapter, User, VoiceMailRetrieval, VirtualLine, AutomatedAttendantVideo, Unknown)
 - **Values:** Distinct Count of Correlation ID
 - **Filter:** Direction = TERMINATING
 - **Purpose:** Identifies where all calls ultimately terminate, distinguishing between human-answered and automated destinations.
 - **Interpretation Note:** Each slice represents *unique call sessions* (distinct Correlation IDs) grouped by the type of recipient on the terminating leg. Using a distinct count of Correlation ID ensures that each call session is counted only once, even if it included multiple legs or redirections. Filtering to Direction = TERMINATING focuses the analysis on the *final receiving side* of the call, rather than intermediary routing steps.

How does it help the project?

This work establishes a clean, consistent dataset and a validated framework for measuring redirection efficiency within Legal Aid’s phone system. As outlined more in the presentation, my analysis helps lead to specific recommendations about reviewing the call routing set up, strengthening intake availability and simplifying redirection paths. My analysis also helps lead to the insight that a high share of calls end in voicemail and the recommendation to optimize voicemail procedures as a focus area.

Issues faced (if any) and attempts to resolve

Issue	Attempt to resolve
Multiple legs can exist under one Correlation ID.	<i>After discovering this, I made a note in my dashboard that the visuals show redirection event volume/duration, not unique calls. Double counting still can reflect the overall routing load. For future work, I plan to aggregate at call-level for unique behavior.</i>
There were blank values in both the ‘Original Reason’ and ‘Redirect Reason’ columns.	<i>I assumed that blank values indicated the call was not redirected but would like to confirm this. I also reviewed related fields such as Answer Indicator and User Type to identify cases where calls may have been redirected even with blank reason fields. For instance, some calls showed “Yes–PostRedirection” in the</i>

	<i>Answer Indicator field, suggesting redirection occurred despite missing reason data. I need clarification on why this is the case. My hypothesis is that this specific case happens when redirection happens automatically through system-level routing or user setting and isn't logged as a redirect reason.</i>
The reference table mapping phone numbers to lines/services was incomplete.	<i>Not used officially in work this time; will need an updated version later to help identify human-connected calls.</i>
Only NoAnswer, FollowMe, Deflection, Unconditional, UserBusy, and Unavailable appeared as redirect reasons in the data. The official documentation lists others (CallQueue, HuntGroup, TimeOfDay, Explicitxxx, Implicitid, Unrecognized).	<i>Need client clarification on why these categories are missing.</i>
The documentation defines call not answered as "the party isn't available to take the call. CF/busy or Voicemail/busy." What does CF stand for here?	<i>Need client clarification on this.</i>

Issues resolved (if any)

See above.

Next steps

- Refine Data Logic and Validation
 - Create logic to aggregate multiple legs under a single Correlation ID to analyze unique calls
 - Verify whether blank Original Reason and Redirect Reason fields truly indicate no redirection or if automatic/system-level redirects are being excluded from logs.
- Enhance Classification
 - Once the phone-number mapping sheet is updated, incorporate it to strengthen classification of Success endpoints in RSR.
- Expand the Dashboard
 - Incorporate a Redirect Success Rate (RSR) visual to monitor redirection efficiency across different redirect types.
 - Introduce visuals like an average duration heatmap by redirect reason, redirect reason success stacked bar chart, etc.

References (Mention if you built up on someone else's work)

- [Column name/Description Documentation](#)
- Professor Kirshna Import Code file for All Calls